Manwe Castro

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Education

Bachelor of Science in Mechanical Engineering — Tempe, AZ
Arizona State University | GPA: 3.61 — *May 2028*
Relevant Coursework: Mechanics of Materials, Structure and Properties of Materials, Thermodynamics, Dynamics, Circuits

Project Experience

Mental Health Robotics System for Isolated Environments — Tempe, AZ
Design and Prototyping Lead — *August 2024 – December 2024*
- Programmed an Arduino Uno to process ultrasonic and photoresistive sensor inputs, controlling drive motors and a servo actuator to enable autonomous user tracking and automated medical dispensing, achieving 95%+ proximity detection reliability, repeatable 15 cm stopping distance, and drawer activation below a 25 Lux light value threshold.
- Modeled a 1.8 cubic foot robotic enclosure in SolidWorks and fabricated the structure from laser-cut balsa and plywood, integrating sensors, motors, display hardware, and battery components into a prototype build, finishing 40% under the $75 budget while maintaining lightweight structural rigidity.
- Executed formal acceptance testing across functional requirements, diagnosing and resolving an electrical power delivery fault to pass sensor, motor, and dispensing tests, validating semi-autonomous navigation and automated medical access.

VR Rapid Prototyping Application — Tempe, AZ
Prototyping Lead | Modeling Team — *March 2025 – December 2025*
- Engineered a virtual reality prototyping environment for Meta Quest headsets using Unity and C#, integrating over 100 interactive 3D assets with simulated physics and collision mechanics to support rapid design across 30+ headsets
- Directed a cross-functional modeling team by establishing standardized Unity workflows and resolving engine-level rendering blockers, optimizing frame rates to mitigate VR motion sickness
- Collaborated with faculty to integrate a 2-week curriculum into the experience and developed in-game analytics scripts to track completion time, errors, and hints, creating a data-driven feedback loop for instructors.
- Authored Gantt charts, risk matrices, and standardized technical reports for a 12-person, multi-semester project, delivering a transition package that enabled future teams to continue development, testing, and classroom deployment.

Shock Absorber Assembly — *August 2025 – December 2025*
- Designed a fully constrained automotive shock absorber assembly in SolidWorks by modeling custom components, integrating McMaster-Carr hardware, and amending dimensions to eliminate physical interferences for dynamic suspension applications.
- Generated fully dimensioned 2D engineering drawings following ASME Y14.5 standards, detailing complex features such as shaft thread specifications, keyways, and spring parameters to provide precise fabrication instructions.
- Authored a comprehensive Bill of Materials and exploded assembly drawing to document custom and standard hardware, establishing a standardized technical reference to streamline part procurement.

Professional Experience

Institute for Digital Inclusion Acceleration — Chandler, AZ
Digital Navigator — *January 2025 – Present*
- Instruct over 300 weekly visitors in hardware and software fundamentals, facilitating thousands of hands-on technology interactions across diverse age groups.
- Develop and deliver structured curriculum for nine distinct software platforms, including Swift, TinkerCAD, and SpheroEDU, standardizing the instructional process for the center.
- Operate and maintain Flashforge 3D printers using Orca Slicer, troubleshooting hardware failures and optimizing print parameters to ensure consistent output quality.

Leadership and Campus Involvement

Society of Hispanic Professional Engineers — Tempe, AZ
Outreach Director — *March 2026 – Present*
- Direct chapter outreach by partnering with local schools and community organizations, prioritizing schools with large Hispanic student populations.
- Plan and coordinate outreach and professional development events of up to 50 attendees including Noche de Ciencias and Engineers Week.
- Manage a 4-member student events committee, 15+ volunteers, and a $500 annual budget while designing 4 hands-on technical workshops throughout the semester that build practical engineering skills for undergraduate members.

Skills

CAD & Design: SolidWorks, Fusion 360, 2D Drafting, MATLAB
Fabrication & Hands-On: FDM 3D Printing, Circuit Wiring. Laser Cutting, Orca Slicer, Automotive Repair
Programming & Controls: Python, Arduino IDE (C++), Unity Game Engine, Git, Sensor Integration
Personal: Violin, Guitar, Sewing, Piano, Running, Jarana, Requinto, Boxing, Rockclimbing, Photography, Salsa